

- Lagrangeve enačbe

• akcija $S = \int L(\vec{r}, \vec{p}, t) dt$, $L = T - V = \frac{1}{2} m \vec{v}^2 - V(\vec{r})$

• 2 variacija akcije $\delta S = 0$ dabiš E-L enačbe:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \vec{p}} \right) - \frac{\partial L}{\partial \vec{r}} = 0$$

- Hamiltonove enačbe

• impuls $\vec{p} = \frac{\partial L}{\partial \vec{v}}$

• Ham. funkcija: $H(\vec{r}, \vec{p}, t) = \vec{p} \vec{v} - L(\vec{r}, \vec{p}, t) = T + V$

$$\dot{\vec{r}} = \frac{\partial H}{\partial \vec{p}} \quad \dot{\vec{p}} = -\frac{\partial H}{\partial \vec{r}}$$

- Lagrangeva funkcija nabitega delca v polju

• Lorentzova sila: $m \vec{\ddot{v}} = e \vec{E} + e \vec{v} \times \vec{B} = -e \nabla \varphi - e \underbrace{\vec{v} \times (\nabla \times \vec{A})}_{\vec{v}(\vec{v} \cdot \vec{A}) - (\vec{v} \cdot \vec{v}) \vec{A}} = -e \nabla \varphi - e \nabla (\vec{v} \cdot \vec{A}) - e \frac{d\vec{A}}{dt}$

$$\Rightarrow \frac{d}{dt} (m \vec{v} + e \vec{A}) = -e (\vec{v} \times \vec{A})$$

$$\frac{d}{dt} \left(\frac{1}{2} m \vec{v}^2 + e \vec{A} \vec{v} \right) = -e (\vec{v} \cdot \vec{v}) \vec{A} \quad \text{Podoben E-L enačbi}$$

$$\Rightarrow L = \frac{1}{2} m \vec{v}^2 - e \varphi + e \vec{v} \cdot \vec{A}$$

- Hamiltonova funkcija nabitega delca v polju

$$H = \vec{p} \vec{v} - L \quad \vec{p} = \frac{\partial L}{\partial \vec{v}} = m \vec{v} + e \vec{A} \Rightarrow \vec{v} = \frac{\vec{p} - e \vec{A}}{m}$$

$$\Rightarrow H = \frac{(\vec{p} - e \vec{A})^2}{2m} - e \varphi + \frac{1}{2} \frac{(\vec{p} - e \vec{A})^2}{m} + e \varphi - e \left(\frac{\vec{p} - e \vec{A}}{m} \right) \cdot \vec{A} = \frac{(\vec{p} - e \vec{A})^2}{2m} - e \varphi + \frac{1}{2} \frac{(\vec{p} - e \vec{A})^2}{m} - e \varphi = \frac{1}{2m} (\vec{p} - e \vec{A})^2 - e \varphi$$

- Schwartzschildova invarianca

• Lagrangeova f. za zvezno porazdelitev nabityh delcev v zun. polju

• kinetični del nos na ravnanje $L_{00} = -e \varphi + e \vec{A} \vec{v}$ (za 1 delec)

$$L_{00} = \int \rho \varphi d^3r + \int \vec{j} \cdot \vec{A} d^3r = \int L_{00} d^3r, \quad \boxed{L_{00} = \rho(\vec{r}, t) \varphi(\vec{r}, t) + \vec{j}(\vec{r}, t) \cdot \vec{A}(\vec{r}, t)} \text{ Sch. inv.}$$

- Lagrangeva funkcija elektromagnetnega polja

(Lag. f. nabityh delcev kot virov polja, ki se nahajajo v zun. EM polju)

$$L = \int L_p d^3r = \int L_p d^3r + \int L_m d^3r$$

↑
Lagrangeva funkcija
za polje

$$L_p(\vec{r}, t) = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \quad (\text{garnung})$$

$$\Rightarrow \boxed{L_p(\vec{r}, t) = \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 - \rho \varphi + \vec{j} \cdot \vec{A}}$$

- E-L in Riemann-Lorentzove enačbe

$$S = \int L(\varphi(\vec{r}, t), \vec{A}(\vec{r}, t)) d^3r dt$$

$$\Rightarrow \frac{1}{dt} \frac{\partial L}{\partial \left(\frac{\partial \varphi}{\partial t} \right)} + \nabla \left(\frac{\partial L}{\partial \nabla \varphi} \right) - \frac{\partial L}{\partial \varphi} = 0 \quad \text{EL enačbe}$$

$$\frac{1}{dt} \frac{\partial L}{\partial \left(\frac{\partial \vec{A}}{\partial t} \right)} + \nabla \left(\frac{\partial L}{\partial \nabla \vec{A}} \right) - \frac{\partial L}{\partial \vec{A}} = 0$$

$$\Rightarrow \boxed{\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\epsilon_0}} \quad \text{Riemann-Lorentzove enačbe}$$

• ki enačbi sta izraz Max. en. na nivoju potencialov